

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)		temporary hard water (1) award (1) for any of following • same volume of soap solution needed (to produce a lather) after boiling as sample C / soft water • same volume of soap solution needed (to produce a lather) after boiling as after adding washing soda • it is soft after boiling / hardness is removed by boiling neutral answers - less soap needed after boiling hardness decreased after boiling			2	2		2
		(ii)		D (1) award (1) for either of following • less soap solution needed (to produce a lather) after boiling but still more than needed for sample C / soft water • boiling did not remove all the hardness / partly softened by boiling • less soap needed after boiling and less again after washing soda			2	2		2

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	(b)			award (1) for similarity in results e.g. (both groups can draw the) same conclusions for samples A, C and D - volumes in the second experiment are approximately three quarters the values in the first experiment for samples A, C and D - results for samples A, C and D are reproducible award (1) for different conclusions drawn for sample B e.g. - first group found sample B to be permanently hard water but the second group found that it contains some temporary hardness and some permanent hardness			2	2		2
	(c)			calcium / magnesium ions <u>swap places with</u> / replaced by / exchanged for sodium ions (2) there must be one reference to <u>ions</u> if not complete answer award (1) for either of following calcium / magnesium ions removed / replaced sodium ions added if no mention of ions anywhere award (1) for calcium / magnesium swaps places with / replaced by / exchanged for sodium do not accept 'displace'	2			2		
	(d)	(i)		$3.2 \times 10^{-3} \text{ mol}$ (2) award (1) for 0.0032		2		2	2	

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		(ii)		moles $\text{MgCl}_2 = \frac{0.384}{95} = 4.04 \times 10^{-3} \text{ mol}$ more Mg²⁺ ions in second sample / magnesium chloride sample therefore it has more hardness (1) accept more moles of magnesium chloride than magnesium sulfate (in the same mass because M_r of magnesium chloride is smaller) ecf possible from part (d) (i)			2	2	1	
				Question 10 total	2	2	8	12	3	6